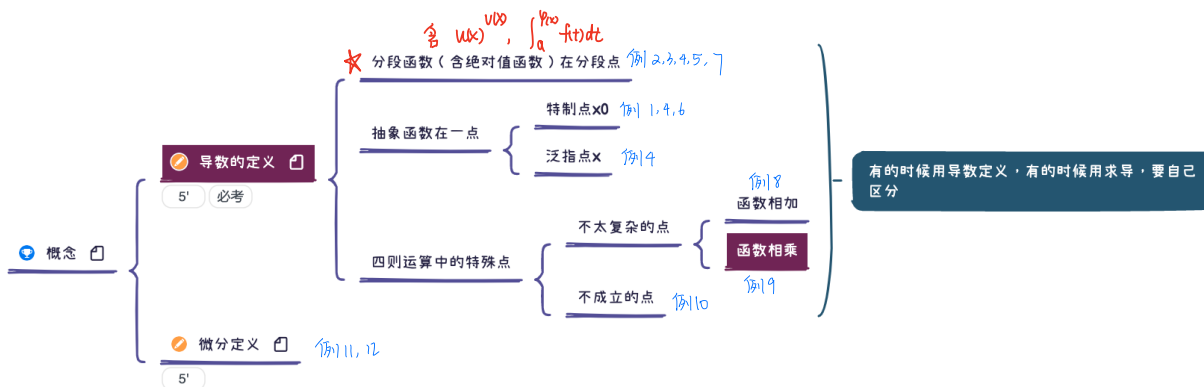


高等数学

一元函数微分学概念



$\star$ 习 3.1, 3.8

例 3.1

例 3.1 设函数 $f(x)$ 在区间 $(-1, 1)$ 内有定义, 且 $\lim_{x \rightarrow 0} f(x) = 0$, 则 ().

- (A) 当 $\lim_{x \rightarrow 0} \frac{f(x)}{\sqrt{|x|}} = 0$ 时, $f(x)$ 在 $x = 0$ 处可导 $f(x) = \begin{cases} x^{\frac{1}{2}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$
- (B) 当 $\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = 0$ 时, $f(x)$ 在 $x = 0$ 处可导
- (C) 当 $f(x)$ 在 $x = 0$ 处可导时, $\lim_{x \rightarrow 0} \frac{f(x)}{\sqrt{|x|}} = 0$
- (D) 当 $f(x)$ 在 $x = 0$ 处可导时, $\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = 0$

(C) 可导 $\rightarrow$ 连续 $\Rightarrow f(0) = A$

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x)}{x} = \lim_{x \rightarrow 0} \left(\frac{x}{\sqrt{|x|}} \right) \cdot A = 0 \Rightarrow (C)$$

$$f(x) = \frac{1}{n}$$

$$1 \leq \frac{f(x)}{x} < \frac{n+1}{n}$$

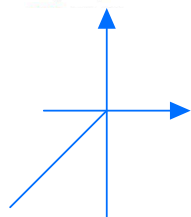
例 3.2

例 3.2 已知函数 $f(x) = \begin{cases} x, & x \leq 0, \\ \frac{1}{n}, & \frac{1}{n+1} < x \leq \frac{1}{n}, n = 1, 2, \dots \end{cases}$, 则 (D).

- (A) $x = 0$ 是 $f(x)$ 的第一类间断点 (B) $x = 0$ 是 $f(x)$ 的第二类间断点
- (C) $f(x)$ 在 $x = 0$ 处连续但不可导 (D) $f(x)$ 在 $x = 0$ 处可导

$$f'_-(0) = 1$$

$$f'_+(0) = \lim_{x \rightarrow 0^+} \frac{f(x) - 0}{x - 0} = \lim_{x \rightarrow 0^+} \left(\frac{f(x)}{x} \right)$$



例 3.3

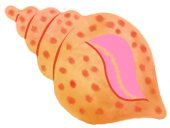
已知函数 $f(x) = \begin{cases} \frac{\int_x^{2x} e^{t^2} dt}{x}, & x \neq 0 \\ a, & x = 0 \end{cases}$, 在 $x = 0$ 处可导, 求

(1) a 的值;

(2) $f'(0)$.

$$(1) \lim_{x \rightarrow 0} \frac{\int_x^{2x} e^{t^2} dt}{x} = \lim_{x \rightarrow 0} \frac{(2x-x) e^{x^2}}{x} = \lim_{x \rightarrow 0} e^{x^2} = 1 = a$$

$$(2) f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{\frac{\int_x^{2x} e^{t^2} dt}{x} - 1}{x} = \lim_{x \rightarrow 0} \frac{\int_x^{2x} e^{t^2} dt - x}{x^2} = \lim_{x \rightarrow 0} \frac{2e^{4x^2} - e^{x^2} - 1}{2x} = \lim_{x \rightarrow 0} \frac{2(e^{4x^2} - 1) - (e^{x^2} - 1)}{2x} = \lim_{x \rightarrow 0} \frac{8x^2 - x^2}{2x} = 0$$



例 3.4

设 $f(x)$ 在 $x = a$ 处连续, $F(x) = f(x) |x - a|$, 则 $f(a) = 0$ 是 $F(x)$ 在 $x = a$ 处可导的 (A).

(A) 充要条件

(B) 充分非必要条件

(C) 必要非充分条件

(D) 既非充分又非必要条件

背

$$(1) \lim_{x \rightarrow a} \frac{F(x) - F(a)}{x - a} = \lim_{x \rightarrow a} \frac{f(x)|x - a|}{x - a} = 0$$

$$(2) \lim_{x \rightarrow a} \frac{F(x) - F(a)}{x - a} = \lim_{x \rightarrow a} \frac{|x - a| f(x)}{x - a} = \lim_{x \rightarrow a} f(x) = 0$$



例 3.5

设有函数

$$f_1(x) = (x^2 - 1) |x^3 + x^2 - 2x - 2|,$$

$$f_2(x) = (x^2 - 1) |x^3 - 2x^2 - x + 2|,$$

$$f_3(x) = (x^2 - 1) |x^3 + 3x^2 - 2x - 6|,$$

将函数 $f_i(x)$ 的不可导点个数记为 $n_i (i = 1, 2, 3)$, 则 (A).

(A) $n_2 < n_1 < n_3$

(B) $n_1 < n_2 < n_3$

(C) $n_3 < n_2 < n_1$

(D) $n_2 < n_3 < n_1$

$$x+1 \sqrt{x^3+x^2-2x-2} \\ \underline{x^3+x^2} \\ x^2-2x-2$$

$$(x+1)(x^2-2)$$

$$\downarrow \text{同号} \quad \downarrow \text{异号} \\ x = \pm \sqrt{2} \text{ 号} \\ \textcircled{2}$$

$$x-1 \sqrt{x^3-2x^2-x+2} \\ \underline{x^3-x^2} \\ -x^2-x+2 \\ \underline{-x^2-x} \\ -2x+2$$

$$(x-1)(x-2)(x+1) \\ \downarrow \quad \downarrow \quad \downarrow \\ \textcircled{1}$$

$$x-\sqrt{2} \sqrt{x^3+3x^2-2x-6} \\ \underline{x^3+\sqrt{2}x^2} \\ x^2-2x-6$$

$$\underline{3\sqrt{2}x^2-2x} \\ 3\sqrt{2}x^2-2x-6 \\ \underline{3\sqrt{2}x^2-6\sqrt{2}x} \\ 3\sqrt{2}-4x-6$$

$$(x-\sqrt{2})(x+\sqrt{2})(x+3) \\ \textcircled{3}$$

例 3.6

例 3.6 设 $f(x)$ 在 $x=0$ 处连续, 且 $\lim_{x \rightarrow 0} \frac{e^{f(x)} - \cos x + \sin x}{x} = 0$, 求 $f(0)$, 并讨论 $f(x)$

在 $x=0$ 处是否可导? 若可导, 求出 $f'(0)$.

$$\textcircled{1} \lim_{x \rightarrow 0} \frac{e^{f(x)} - \cos x + \sin x}{x} = 0 \Rightarrow \frac{e^{f(x)} - \cos x + \sin x}{x} = 0 + 0$$

$$\Rightarrow e^{f(x)} = \alpha \cdot x + \cos x - \sin x$$

$$f(x) = \ln(\alpha x + \cos x - \sin x)$$

$$f(x) - f(0) \stackrel{\Delta}{=} \lim_{x \rightarrow 0} f(x) = 0 \quad \star \ln u \sim u-1, u \rightarrow 1$$

$$\textcircled{2} f(0) \stackrel{\Delta}{=} \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{\ln(\alpha x + \cos x - \sin x)}{x} = \lim_{x \rightarrow 0} \frac{\alpha x + \cos x - \sin x - 1}{x} = -1$$

例 3.7

例 3.7 设 $f(x) = \begin{cases} \frac{1 - \cos x}{\sqrt{x}}, & x > 0, \\ x^2 g(x), & x \leq 0, \end{cases}$ 其中 $g(x)$ 是有界函数, 则 $f(x)$ 在 $x=0$

处 (D).

(A) 极限不存在

(B) 极限存在, 但不连续

(C) 连续, 但不可导

(D) 可导

$$\textcircled{1} \lim_{x \rightarrow 0^+} \frac{1 - \cos x}{\sqrt{x}} = \lim_{x \rightarrow 0^+} \frac{\frac{1}{2} x^2}{x^{\frac{1}{2}}} = 0 \quad \left. \begin{array}{l} \lim_{x \rightarrow 0^-} x^2 g(x) = 0 \end{array} \right\} \text{连续}$$

$$\textcircled{2} \lim_{x \rightarrow 0^+} \frac{\frac{1 - \cos x}{\sqrt{x}} - 0}{x - 0} = 0 \quad \left. \begin{array}{l} \lim_{x \rightarrow 0^-} \frac{x^2 g(x) - 0}{x - 0} = 0 \end{array} \right\} \text{可导}$$

例 3.8

例 3.8 设 $f(x) = \sqrt{\frac{(1+x)\sqrt{x}}{e^{x-1}}} + \arcsin \frac{1-x}{\sqrt{1+x^2}}$, 求 $f'(1)$.

$$\textcircled{1} \text{ 令 } u = \sqrt{\frac{(1+x)\sqrt{x}}{e^{x-1}}}$$

$$\ln u = \frac{1}{2} (\ln(1+x) + \frac{1}{2} \ln x - (x-1))$$

$$\frac{1}{u} \cdot u' = \frac{1}{2} \left(\frac{1}{1+x} + \frac{1}{2x} - 1 \right)$$

$$u'|_1 = u|_{x=1} \cdot \frac{1}{2} \left(\frac{1}{1+1} + \frac{1}{2 \cdot 1} - 1 \right) \Big|_{x=1} = 0$$

$$\textcircled{2} \arcsin \frac{1-x}{\sqrt{1+x^2}}$$

$$v'(1) = \lim_{x \rightarrow 1} \frac{\arcsin \frac{1-x}{\sqrt{1+x^2}}}{x-1} = \lim_{x \rightarrow 1} \frac{-\frac{1}{\sqrt{1+x^2}}}{1+x^2} = -\frac{1}{12}$$

$$0 - \frac{1}{12} = -\frac{1}{12}$$

例 3.9

例 3.9 设函数 $f(x) = (e^x - 1)(e^x - 2) \cdots (e^x - n)$, 其中 n 为正整数, 则 $f'(0) = ()$.

- (A) $(-1)^{n-1}(n-1)!$ (B) $(-1)^n(n-1)!$
(C) $(-1)^{n-1}n!$ (D) $(-1)^nn!$

$$\begin{aligned} f'(0) &= e^0 (1-2)(1-3) \cdots (1-n) \\ &= (-1)^{n-1} (n-1)! \quad (A) \end{aligned}$$

例 3.10

例 3.10 设 $f(x) = \sqrt[3]{x} \sin x$, 求 $f'(x)$.

$$f'(x) = \frac{2}{3}x^{-\frac{1}{3}} \sin x + x^{\frac{2}{3}} \cos x \quad X$$

$x \neq 0$ 时

$$x=0 \text{ 时 } f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{x^{\frac{2}{3}} \sin x}{x} = 0.$$

例 3.11

例 3.11 设 $y = \arctan \frac{2x}{1+x^2}$, 求 dy .

$$\begin{aligned} \left(\begin{aligned} \Delta y &= f(x_0 + \Delta x) - f(x_0) \\ dy &= A \Delta x \end{aligned} \right) \\ dy &= f'(x) \cdot dx = \frac{\frac{2(1+x^2) - 2x \cdot 2x}{(1+x^2)^2}}{1 + \frac{4x^2}{(1+x^2)^2}} dx \end{aligned}$$

例 3.12

例 3.12 求由方程 $2y - x = (x - y) \ln(x - y)$ 所确定的函数 $y = y(x)$ 的微分 dy .

$$\begin{aligned} 2dy - dx &= (dx - dy) \ln(x - y) + \frac{(x - y)}{x - y} (dx - dy) \\ \Rightarrow dy &= \frac{x}{2x - y} dx \end{aligned}$$

习 3.1

3.1 设 $f(x)$ 可导, $F(x) = f(x)(1 + |\sin x|)$, 若使 $F(x)$ 在 $x = 0$ 处可导, 则必有

- (A) $f(0) = 0$ (B) $f'(0) = 0$
(C) $f(0) + f'(0) = 0$ (D) $f(0) - f'(0) = 0$

$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{f(x)(1 + |\sin x|) - f(0)}{x - 0} &= \lim_{x \rightarrow 0^+} f'(x)(1 + |\sin x|) + f(x) \cdot \cos x = \underline{f'(0) + f(0)} \\ \lim_{x \rightarrow 0^-} \frac{f(x)(1 - |\sin x|) - f(0)}{x - 0} &= f'(0) - f(0) \end{aligned}$$

习3.2

3.2 设 $f(x)$ 在 x_0 处二阶可导, 且 $f'(x_0) < 0, f''(x_0) < 0, \Delta x > 0$, 记 $\Delta y = f(x_0 + \Delta x) - f(x_0), dy = f'(x_0)\Delta x$, 则 (A).

(A) $\Delta y < dy < 0$ (B) $\Delta y > dy > 0$ (C) $dy > \Delta y > 0$ (D) $dy < \Delta y < 0$



习3.3

3.3 设 $f(x)$ 在 $x=1$ 处可导, $f'(1)=1$, 求 $\lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x^{10} - 1}$.

$$f'(1) = \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1}$$

$$\lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x^{10} - 1} = \lim_{x \rightarrow 1} \frac{f'(1) \cdot (x-1)}{x^{10} - 1} = \lim_{x \rightarrow 1} \frac{x-1}{x^{10} - 1} \stackrel{x=t}{=} \lim_{t \rightarrow 1} \frac{t}{t^{10} - 1} = \lim_{t \rightarrow 1} \frac{1}{10(t-1)^9} = \frac{1}{10}$$

习3.4

3.4 设函数 $f(x)$ 在 $(-\infty, +\infty)$ 上有定义, 且对于任意 $x, y \in (-\infty, +\infty)$, 不等式 $|f(x) - f(y)| \leq M|x - y|^{1+\alpha}$ (其中 M, α 均为正常数) 恒成立. 证明: 对于任意 $x \in (-\infty, +\infty), f(x) \equiv C$ (C 为常数).

$$\text{取 } y = x + \Delta x$$

$$0 \leq |f(y) - f(x)| = \left| \frac{f(x + \Delta x) - f(x)}{\Delta x} \right| \leq M|\Delta x|^\alpha$$

$$\lim_{\Delta x \rightarrow 0} |\Delta x|^\alpha = 0 \Rightarrow f'(x) = 0 \Rightarrow f(x) \equiv C$$

习3.5

3.5 设 $f''(1)$ 存在, 且 $\lim_{x \rightarrow 1} \frac{f(x)}{x-1} = 0$. 记 $\varphi(x) = \int_0^1 f[1 + (x-1)t] dt$, 求 $\varphi(x)$ 在 $x=1$ 的某个邻域内的导数, 并讨论 $\varphi'(x)$ 在 $x=1$ 处的连续性.

$$\textcircled{1} \lim_{x \rightarrow 1} \frac{f(x) - 0}{x - 1} = 0 \Rightarrow f(1) = 0 \text{ 且 } f'(1) = 0$$

$$f'(1) = \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} = \lim_{x \rightarrow 1} \frac{f(x)}{x - 1} = A$$

$$\int_0^1 f(x) dx \quad (1-0)f(1)$$

$$\textcircled{2} \varphi(x) = \int_0^1 f[1 + (x-1)t] dt, \quad \varphi(1) = \int_0^1 f(1) dt = 0$$

$$\varphi'(1) = \lim_{x \rightarrow 1} \frac{\varphi(x) - \varphi(1)}{x - 1} = \lim_{x \rightarrow 1} \frac{\varphi(x)}{x - 1} = \lim_{x \rightarrow 1} \frac{\int_0^1 f[1 + (x-1)t] dt}{x - 1} = \lim_{u \rightarrow 0} \frac{\int_0^1 f[1 + ut] dt}{u}$$

$$\text{令 } (x-1)t = u, dt = \frac{1}{x-1} du, \quad \begin{matrix} (x-1)t=1 \\ u=x-1 \end{matrix} = \lim_{u \rightarrow 0} \frac{(1-0)f(1)}{u}$$

$$\textcircled{2} \varphi(x) = \int_0^{x-1} \frac{1}{x-1} f(1+u) du = \frac{f(x) - f(1)}{x-1} = \frac{f(x)}{x-1}, \quad x \neq 1$$

$$\varphi(1) = 0, \quad \varphi'(x) = \frac{f(x)(x-1) - f(x)}{(x-1)^2} = \frac{f'(x)}{x-1} - \frac{f(x)}{(x-1)^2}, \quad (x \neq 1)$$

$$\varphi'(1) = \lim_{x \rightarrow 1} \frac{\varphi(x)}{x-1} = \lim_{x \rightarrow 1} \frac{f(x)}{(x-1)^2} = \lim_{x \rightarrow 1} \frac{f'(x)}{2(x-1)} = \frac{1}{2} f''(1)$$

$$\lim_{x \rightarrow 1} \varphi'(x) = f''(1) - \frac{1}{2} f''(1) = \frac{1}{2} f''(1)$$

$\Rightarrow$ 连续

习3.6

3.6 讨论函数 $f(x) = \begin{cases} \frac{x}{1+e^{\frac{1}{x}}}, & x \neq 0, \\ 0, & x = 0 \end{cases}$ 在 $x=0$ 处是否可导.

$$\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{\frac{x}{1+e^{\frac{1}{x}}}}{x} = \lim_{x \rightarrow 0} \frac{1}{1+e^{\frac{1}{x}}}$$

$$\lim_{x \rightarrow 0^+} \frac{1}{1+e^{\frac{1}{x}}} \neq \lim_{x \rightarrow 0^-} \frac{1}{1+e^{\frac{1}{x}}}, \text{ 不可导}$$

习3.7

3.7 设 $f(x) = \frac{(x-1)(x-2)\cdots(x-n)}{(x+1)(x+2)\cdots(x+n)}$, 求 $f'(1)$.

$$f'(1) = \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} = \lim_{x \rightarrow 1} \frac{(x-2)(x-3)\cdots(x-n)}{(x+1)(x+2)\cdots(x+n)} = \frac{(-1)(-2)\cdots(-n+1)}{(n+1)(n+2)\cdots(n+n)} = \frac{(-1)^{n-1}(n-1)!}{(n+1)!} = \frac{(-1)^{n-1}}{n(n+1)}$$

习3.8

3.8 设 $f(x) = a_1 \sin x + a_2 \sin 2x + \cdots + a_n \sin nx$, 并且 $|f(x)| \leq |\sin x|$, 求证:
 $|a_1 + 2a_2 + \cdots + na_n| \leq 1$.

$$f'(x) = a_1 \cos x + 2a_2 \cos 2x + 3a_3 \cos 3x + \cdots + na_n \cos nx$$

$$f'(0) = a_1 + 2a_2 + \cdots + na_n$$

$$\text{当 } x \neq 0 \text{ 时 } |f(x)| \leq |\sin x|, \quad f(0) = \sin 0 \Rightarrow |f'(x)| \leq |\sin x|' = \cos x \leq 1$$

$$\text{当 } x=0 \text{ 时 } f'(x) =$$

$$|f'(0)| = \left| \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} \right| = \lim_{x \rightarrow 0} \left| \frac{f(x)}{x} \right| \leq \lim_{x \rightarrow 0} \left| \frac{\sin x}{x} \right| = 1$$

$$\text{故有 } a_1 + 2a_2 + \cdots + na_n \leq 1$$



习3.9

3.9 设 $f(x) = \begin{cases} e^{ax}, & x \leq 0, \\ (x-b)^3, & x > 0 \end{cases}$ 在 $x=0$ 处可导.

(1) 求常数 a, b 的值;

(2) 求 $f(x)$ 在 $x=0$ 处的微分.

$$(1) \lim_{x \rightarrow 0^+} f(x) = -b^3 = 1 = f(0) \Rightarrow b = -1$$

$$\left. \begin{aligned} \lim_{x \rightarrow 0^+} f'(x) &= a \\ \lim_{x \rightarrow 0^-} f'(x) &= 3 \end{aligned} \right\} \Rightarrow a = 3$$

$$f(x) = \begin{cases} e^{3x}, & x \leq 0 \\ (x+1)^3, & x > 0 \end{cases}$$

$$(2) df(0) = f'(0) \cdot dx = 3dx$$

习3.10

$=t$

3.10 设 $x^y = y^x$, 其中 y 是 x 的函数, 求 $\frac{dy}{dx}$.

x^a

$$x^y = y^x \Rightarrow F(x, y) = x^y - y^x = 0$$

$$\frac{dy}{dx} = -\frac{F'_x}{F'_y} = -\frac{-(yx^{y-1} - y^x \ln y)}{x^y \ln x - xy^{y-1}} = \frac{y(x^{y-1} - y^{x-1} \ln y)}{x(y^{y-1} - x^{y-1} \ln x)} = \frac{y(\frac{1}{x} - \frac{1}{y} \ln y)}{x(\frac{1}{y} - \frac{1}{x} \ln x)} = \frac{y(y - x \ln y)}{x(x - y \ln x)}$$

习3.1.1

1. 设函数 $f(x) = \begin{cases} \frac{2^x - e^{-x} + x}{\arctan x}, & x \neq 0, \\ a, & x = 0 \end{cases}$ 在 $x = 0$ 处连续, 则 $f(x)$ 在 $x = 0$ 处 ().

(A) 可导, 且 $f'(0) = \frac{1}{2}(\ln^2 2 + 1)$ (B) 可导, 且 $f'(0) = \frac{1}{2}(\ln^2 2 - 1)$

(C) 不可导 (D) 是否可导与 a 的取值有关

$$f'(x) = \lim_{x \rightarrow 0} \frac{\frac{2^x - e^{-x} + x}{\arctan x} - a}{x - 0} = \lim_{x \rightarrow 0} \frac{(2^x \ln 2 + e^{-x} + 1) \arctan x - (2^x - e^{-x} + x) \frac{1}{1+x^2}}{(\arctan x)^2}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{2+\ln 2}{1+x^2} - (2^x \ln 2 + e^{-x} + 1)}{2x} = \lim_{x \rightarrow 0} \frac{-2^x \ln^2 2 - e^{-x}}{2} = \frac{-\ln^2 2 - 1}{2}$$

习2.1.2

2. 设函数 $f(x) = \begin{cases} \frac{e^x - 1}{x}, & x > 0, \\ x^2 g(x), & x \leq 0, \end{cases}$ 其中 $g(x)$ 为有界函数, 则在 $x = 0$ 处, $f(x)$ ().

(A) 极限不存在

(B) 极限存在但不连续

(C) 连续但不可导

(D) 可导

$$\lim_{x \rightarrow 0^+} \frac{e^x - 1}{x} = 0 \quad \lim_{x \rightarrow 0} x^2 g(x) = 0$$

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} - A = 1 \quad \lim_{x \rightarrow 0} \frac{x^2 g(x)}{x} = 0$$

习2.1.3

3. 设 $f(x) = \int_0^1 \ln \sqrt{x^2 + y^2} dy, 0 \leq x \leq 1$, 则 $f'_+(0) = ()$.

(A) $-\frac{\pi}{2}$

(B) $\frac{\pi}{2}$

(C) $-\pi$

(D) π

$$x^2 + y^2 = t^2 \Rightarrow t = \sqrt{x^2 + y^2}, \quad y = \sqrt{t^2 - x^2} \quad dy = \frac{2t}{2\sqrt{t^2 - x^2}} = \frac{t}{\sqrt{t^2 - x^2}}$$

$$f(x) = \int_0^1 \ln \sqrt{x^2 + y^2} dy = \ln \sqrt{x^2 + y^2} \cdot y \Big|_0^1 - \int_0^1 \frac{y}{\sqrt{x^2 + y^2}} d\sqrt{x^2 + y^2}$$

$$= \frac{1}{2} \ln(1+x^2) - \int_0^1 \frac{y^2}{x^2 + y^2} dy = \frac{1}{2} \ln(1+x^2) - x \int_0^{\frac{1}{x}} \frac{(\frac{t}{x})^2}{1 + (\frac{t}{x})^2} d\frac{y}{x} = \frac{1}{2} \ln(1+x^2) - x \int_0^{\frac{1}{x}} \frac{t^2 dt}{1+t^2}$$

$$= \frac{1}{2} \ln(1+x^2) - 1 + x \int_0^{\frac{1}{x}} \frac{1}{1+t^2} dt = \frac{1}{2} \ln(1+x^2) - 1 + x \arctan \frac{1}{x}$$

$$f'(x) = \frac{x}{1+x^2} + \arctan \frac{1}{x} + \frac{x}{1+x^2}$$

$$f'_+(0) = \frac{\pi}{2} \quad (B)$$

72.1.5

5. 设函数 $f(x)$ 可导, 且 $y = f(x^3)$. 当自变量 x 在 $x = -1$ 处取得增量 $\Delta x = -0.1$ 时, 相应的函数增量 Δy 的线性主部为 0.3, 则 $f'(-1) = ()$.

- (A) -1 (B) 0.1 (C) 1 (D) 0.3

$$y = f(x^3) \quad x = -1 \quad \Delta x = -0.1 \quad dy = f'(x) \cdot \underbrace{3x^2}_{(3)} dx$$

$$0.3 = 3 \cdot f'(-1) \cdot (-0.1) \Rightarrow f'(-1) = -1$$

(A)

72.1.6

6. 若 $f(x) = e^{10x}(x+1)(x+2)\cdots(x+10)$, 则 $f'(0) = \underline{\quad\quad\quad}$.

$$f'(x) = x' \cdot g(x) + x \cdot g'(x)$$

$$f'(0) = g(0) + 0 = 1 \cdot 2 \cdot 3 \cdots 10 = 10!$$

72.1.4

4. 设函数 $f(x)$ 是定义在 $(-1, 1)$ 内的奇函数, 且 $\lim_{x \rightarrow 0^+} \frac{f(x)}{x} = a \neq 0$, 则 $f(x)$ 在 $x = 0$ 处的导数为 ().

- (A) a (B) $-a$ (C) 0 (D) 不存在

$$\lim_{x \rightarrow 0^-} \frac{f(x)}{x} = \lim_{t \rightarrow 0^+} \frac{-f(t)}{-t} = a \quad (A)$$

72.1.7

7. 设对任意 x , $f(x)$ 满足 $f(x+1) = 2f(x)$, 且 $f(0) = 1, f'(0) = C$ (常数), 则 $f'(1) =$

$$f(x+1) = 2f(x) \quad f(1) = 2f(0) = 2$$

$$\lim_{x \rightarrow 1} \frac{f(x) - 2}{x - 1} = \lim_{t \rightarrow 0} \frac{f(t+1) - 2}{t} = \lim_{t \rightarrow 0} \frac{2f(t) - 2}{t} = 2 \lim_{t \rightarrow 0} \frac{f(t) - f(0)}{t} = 2C$$

72.1.8

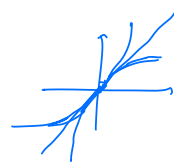
8. 已知 $f(x) = \sqrt{1+x} + \arcsin \frac{1-x}{1+x^2}$, 则 $f'(1) = \underline{\quad\quad\quad}$.

$$f_1(x) = \sqrt{1+x} \quad f_2(x) = \arcsin \frac{1-x}{1+x^2}$$

$$f_1'(x) = \frac{1}{2\sqrt{1+x}} \quad f_2'(x) = \frac{-(1+x^2) - (1-x) \cdot 2x}{(1+x^2)^2}$$

$$f_2'(x) = \frac{-1-x^2-2x+2x^2}{(1+x^2)^2} = \frac{x^2-2x-1}{(1+x^2)^2}$$

$$f'(1) = \frac{\sqrt{2}}{4} + \frac{-2}{4} = \frac{\sqrt{2}-2}{4}$$



72.1.9

9. 设 $f(x)$ 可导, $f(0) = f'(0) = 1$, 则 $\lim_{x \rightarrow 0} \frac{f(x)e^x - 1}{f(x)\cos x - 1} = \underline{\quad\quad\quad}$.

$$\lim_{x \rightarrow 0} \frac{f'(x)e^x + f(x)e^x}{f'(x)\cos x - f(x)\sin x} = \frac{2}{1} = 2$$

7.1.10

10. 设 $y = f(x)$ 由方程 $\sin(xy) + \ln y - x = 1$ 确定, 则 $\lim_{n \rightarrow \infty} n \left[f\left(\frac{2}{n}\right) - e \right] = \underline{\hspace{2cm}}$.

$$\begin{aligned} \lim_{n \rightarrow \infty} n \left[f\left(\frac{2}{n}\right) - e \right] &= \lim_{x \rightarrow 0} 2 \frac{f(2x) - e}{2x} = \lim_{t \rightarrow 0} 2 \frac{f(t) - e}{t} = \lim_{t \rightarrow 0} 2 f'(t) \\ F(x, y) &= \sin(xy) + \ln y - x - 1 = 0 \quad x=0 \text{ 时: } \ln y = 1 \Rightarrow y = e \\ \frac{dy}{dx} &= -\frac{F'_x}{F'_y} = -\frac{y \cos xy - 1}{x \cos xy + \frac{1}{y}} \\ \Rightarrow 2f'(0) &= -2 \frac{e \cdot -1}{\frac{1}{e}} = 2e(1-e) \end{aligned}$$

7.1.11

11. 函数 $f(x)$ 在 $(-\infty, +\infty)$ 内有定义, 在区间 $[0, 2]$ 上, $f(x) = x(x^2 - 4)$. 设对任意的 x 都满足 $f(x) = kf(x+2)$, 其中 k 为常数.

(1) 写出 $f(x)$ 在 $[-2, 0]$ 上的表达式;

(2) 问 k 为何值时, $f(x)$ 在 $x = 0$ 处可导?

$$\begin{aligned} (1) \text{ 设 } x \in [-2, 0] \text{ 则 } -x \in [0, 2], \quad x+2 \in [0, 2] \\ f(-x) &= (-x)(x^2 - 4) \\ \Delta f(x) &= kf(x+2) = k(x+2)(x^2 + 4x) \\ (2) f(x) &= \begin{cases} k(x+2)(x+4)x = k(x^3 + 6x^2 + 8x), & x \in [-2, 0] \\ (x-2)(x+2)x = x^3 - 4x, & x \in [0, 2] \end{cases} \\ \Rightarrow -4 &= 8k \quad k = -\frac{1}{2} \end{aligned}$$

7.1.12

12. 设函数 $y = f(x) = \begin{cases} 1 - 2x^2, & x < -1, \\ x^3, & -1 \leq x \leq 2, \\ 12x - 16, & x > 2. \end{cases}$

(1) 写出 $f(x)$ 的反函数 $g(x)$ 的表达式;

(2) 讨论 $g(x)$ 是否有不可导点, 若有, 指出这些点.

$$\begin{aligned} (1) \begin{cases} y = 1 - 2x^2, & x < -1 \Rightarrow x = -\sqrt{\frac{1-y}{2}}, & y < -1 \Rightarrow g(x) = \sqrt{\frac{1-x}{2}}, & x < -1 \\ y = x^3, & -1 \leq x \leq 2 \Rightarrow x = y^{\frac{1}{3}}, & -1 \leq y \leq 8 \Rightarrow g(x) = \sqrt[3]{x}, & -1 \leq x \leq 8 \\ y = 12x - 16, & x > 2 \Rightarrow x = \frac{y+16}{12}, & 8 \leq y \end{cases} \end{aligned}$$

$$(2) \quad g'(x) = \begin{cases} \frac{1}{2} \sqrt{\frac{2}{1-x}} \cdot (-\frac{1}{2}) = -\frac{1}{4} \sqrt{\frac{2}{1-x}}, & x \in (-\infty, -1) \\ \frac{1}{3} x^{-\frac{2}{3}}, & x \in (-1, 8) \\ \frac{1}{12}, & x \in (8, +\infty) \end{cases}$$

$$\lim_{x \rightarrow -1^-} g'(x) = -\frac{1}{4} \quad \lim_{x \rightarrow -1^+} \frac{1}{3} x^{-\frac{2}{3}} = \frac{1}{3} \quad \text{不可导}$$

& $x=0$ 不可导 *

7d.1.13

13. 设 $f(x)$ 在 $(-\frac{\pi}{2a}, \frac{\pi}{2a})$ ($a > 0$) 内有定义, 且 $f'(0) = a$, 又对任意的 $x, y, x+y \in (-\frac{\pi}{2a}, \frac{\pi}{2a})$,

有 $f(x+y) = \frac{f(x)+f(y)}{1-f(x)f(y)}$, 求 $f(x)$.

$$f(0) = \frac{2f(0)}{1-f(0)^2} \Rightarrow f(0) = 0 \Rightarrow \lim_{x \rightarrow 0} \frac{f(x)}{x} = a$$

$$\begin{aligned} f(2x) &= \frac{2f(x)}{1-f(x)^2} \quad \lim_{x \rightarrow 0} \frac{f(2x)}{2x} = a = \lim_{x \rightarrow 0} \frac{\frac{2f(x)}{1-f(x)^2}}{2x-0} = \lim_{x \rightarrow 0} \frac{2f(x)[1-f(x)^2] + 2f(x)^2 \cdot f'(x)}{(1-f(x)^2)^2} \\ &= \lim_{x \rightarrow 0} f'(x) + 2f(x)^2 f'(x) = a \end{aligned}$$

7d.1.14

14. 设 $f(x)$ 在 $(-\infty, +\infty)$ 内有定义, 且对任意的 $x, x_1, x_2 \in (-\infty, +\infty)$, 有

$$f(x_1 + x_2) = f(x_1) \cdot f(x_2), f(x) = 1 + xg(x),$$

其中 $\lim_{x \rightarrow 0} g(x) = 1$. 证明: $f(x)$ 在 $(-\infty, +\infty)$ 内处处可导.

$$\lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{[f(x) \cdot f(\Delta x)] - f(x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\Delta x g(\Delta x) \cdot f(x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} g(\Delta x) \cdot f(x) = f(x)$$

$$\therefore f(x) \text{ 在 } (-\infty, +\infty) \text{ 内有定义} \Rightarrow \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} \text{ 有定义}$$